



Group transfer entropy with an application to cryptocurrencies

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HIGHLIGHTS

- Group transfer entropy is able to quantify nonlinear dependencies within a system.
- A suitable bootstrap procedure allows inference of Group transfer entropy estimates.
- The linkages between different cryptocurrencies are mostly nonlinear.
- Based on Group transfer entropy Bitcoin is not the dominating cryptocurrency which leads the information process.

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ABSTRACT

The detection of informational leadership is a core issue in financial market microstructure. We use effective group transfer entropy (EGTE) as a measure for the predictability of a stochastic process using lagged observations on multiple related processes within the same system. We propose an appropriate bootstrap to derive confidence bounds and show by means of a simulation study that standard linear approaches in economics and finance, such as vector autoregressions and Granger-causality tests, are not well suited to detect information transfer. We empirically examine the markets for cryptocurrencies using intraday data and reveal that the dependencies are mostly of nonlinear nature, highlighting the applicability of EGTE in the context of this new financial product.

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1. Introduction

The measurement of information transfer between different time series is of interest in many disciplines. With the development of transfer entropy in 2000 [1], researchers were equipped with a tool which stands out for its universal applicability. In recent years transfer entropy has been applied across a variety of research areas, such as thermodynamics [2], econophysics [3–5], neuroscience [6–8], biomedical engineering [9,10], and ecological modeling [11].

In economics and finance, the quantification of information transfer commonly relies on measures that have been derived from subject-specific assumptions and restrictions concerning the underlying stochastic processes or theoretical models. The most commonly used metric is the information share of Hasbrouck [12] or its recent refinements by [13,14], or [15]. The measure provides a decomposition of the variance of the common efficient price innovations which is interpreted as information flow between the considered time series. Such a measure is of particular interest in a setting where one security is traded at multiple trading venues [e.g. 16] or in the context of similar assets linked by the law of one price (e.g. spot and future as in [17]). The methodology, however, requires that the data are cointegrated and can therefore only

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be applied in a limited number of settings. Transfer entropy offers an attractive tool to overcome this limitation imposed by the data requirement and the additional limitation to linear models such as (cointegrated) vector autoregressions (VARs) and Granger-causality [18].

According to Hasbrouck [12, p. 1175], price discovery whereby new information is incorporated into security prices is “the most important product[s] of a security market”. The information leader can hope to attract the largest number of traders which has a positive impact on all aspects of market liquidity. Hence, knowing which market is the leading market, is not only of interest to the trading platforms, but also to investors and regulators. In situations where many similar financial products are traded, it might be of interest which of these is the informational leader. However, as the law of one price cannot be claimed to hold if different assets are considered, cointegration cannot be established. Resorting to a linear VAR framework, tests of Granger-causality are therefore the main tool. The latter still allow to establish informational leadership in the sense that the leader should Granger-cause the other entities while the respective asset itself is not predictable.

In this article, we propose to use *effective group transfer entropy* (EGTE) to measure to which extent a random variable is affected by a group of other random variables. We show that this measure can unveil information transfer beyond linear autoregressive models and propose a suitable time series block-bootstrap procedure for EGTE, which allows to conduct inference. Although transfer entropy and Granger-causality are equivalent for Gaussian variables [19], our simulation study shows that Granger-causality might lead to wrong conclusions, if the underlying processes are related in a nonlinear way. Additionally, we show that effective group transfer entropy is not impacted by structural breaks, which are often an issue in economic and financial time series, caused by regulatory changes or periods of economic distress.

We use a sample of four cryptocurrencies (Bitcoin, Ethereum, Litecoin, and Ripple) to illustrate the use of EGTE. While the efficiency of the Bitcoin market has been extensively studied [20–23], the interdependencies between different cryptocurrencies over time have rarely been analyzed, most likely, because standard financial models and tools are not well suited for application to these cryptocurrencies. However, the fact that Bitcoin is much more prominent (also because of the existence of futures) might suggest that this currency plays a dominant role in the cryptocurrency market which might lead to informational leadership in the price formation process of other cryptocurrencies as well. Using EGTE we show that information transfer actually is mostly nonlinear. Consequently, standard methods, such as Granger-causality tests based on a VAR, are too restrictive to unveil the full nature of dependencies among these new financial products. This result emphasizes the importance of interdisciplinary research areas, such as econophysics, as it might overcome the limitation of standard methods in economics and finance and encourage the use of more flexible measures such as transfer entropy. Furthermore, a clearly dominant position of Bitcoin is not found.

The article proceeds as follows. Section 2 presents group transfer entropy and the bootstrap method. Section 3 illustrates the properties of EGTE in a controlled simulation environment. Section 4 presents our application of EGTE to a set of cryptocurrencies and compares the results to a VAR analysis. Finally, Section 5 concludes.

2. GTE estimation and inference

2.1. Group transfer entropy

Financial or economic systems often involve more than two variables. While commonly feedback effects between different time series might exist, it is often sufficient to determine the extent to which each time series is dependent on the past states of the system to identify the informational leader. We therefore use a transfer entropy based measure to quantify the dependencies of one time series on past values of a group of other time series.

Let X_t be a set of n stochastic processes $\{X_{1,t}, \{X_{2,t}, \dots, \{X_{n,t}\}$ which can individually be described by a Markov process of order p satisfying the Markov property

$$P(X_{i,t}|X_{i,t-1}, \dots, X_{i,t-p}, X_{i,t-p-1} \dots X_{i,1}) = P(X_{i,t}|X_{i,t-1}, \dots, X_{i,t-p}). \quad (1)$$

Denote by Y_t the subset of X_t excluding $X_{i,t}$, i.e. $Y_t = \{X_t \setminus X_{i,t}\}$. Based on [1], we measure information transfer from the group of processes Y_t to $X_{i,t}$ using Group Transfer Entropy (GTE) calculated as

$$\text{GTE}_{Y_t \rightarrow X_{i,t}(k,l)} = \sum p(x_{i,t+1}, x_{i,t}^{(k)}, y_t^{(l)}) \log \frac{p(x_{i,t+1}|x_{i,t}^{(k)}, y_t^{(l)})}{p(x_{i,t+1}|x_{i,t}^{(k)})}, \quad (2)$$

where $\log$ denotes the base 2 logarithm and $x^{(k)}$ ($y^{(l)}$) indicates a Markov process of order k (l). GTE is readily adapted to any number of variables in the conditioning group. For implementation of the measure in Eq. (2), the conditional probabilities are replaced by unconditional probabilities using Bayes' law and the probabilities are estimated by empirical frequencies.

As economic or financial data are realizations of continuous processes, while GTE requires symbolic input, discretization of the data is required [24]. Following previous studies using financial data [cp. 3,25], we propose to apply a partitioning into three bins. A higher number of bins is possible, but requires a considerably greater amount of data. To obtain the symbolically encoded process $\{S_t\}$, a stationary time series $\{X_t\}$ is partitioned as follows:

$$s_t = \begin{cases} A & \text{for } x_t \leq q_1 \\ B & \text{for } q_1 < x_t < q_2 \\ C & \text{for } x_t \geq q_2 \end{cases} \quad (3)$$

The symbolic encoding in Eq. (3) replaces each value x_t in the observed time series by a corresponding symbol (A, B, C). Financial return data commonly exhibit leptokurtic distributions, where the tail observations are of main interest, as they correspond to extremely low or high returns, associated with major information events. In addition multivariate distributions often exhibit tail dependence, where the correlation between financial returns in the tails of the distribution exceeds the correlation in the center [26]. Therefore q_1 and q_2 may be chosen by the empirical quantiles, such that the bins correspond to the tails (symbols A and C) and the center (symbol B) of the distribution. A useful choice is to set $q_1 = x_{[0.05]}$ and $q_2 = x_{[0.95]}$. Doing so, the quantiles are dependent on the empirical distribution of each individual time series, while the number of observations in a particular bin remains the same across all processes. GTE is then calculated based on the relative frequency distribution of the discretized data.

2.2. Effective group transfer entropy

We follow [27] and adapt their shuffling procedure to derive effective GTE estimates. We keep the dependent variable X to conserve its univariate time series properties, but draw random tuples y_t from the group of variables Y in order to destroy the dependencies with X , while keeping the dependencies among the Y variables. We repeat the shuffling 100 times and use the average over these repetitions as an estimator for the bias, which is subtracted from GTE in Eq. (2) to derive effective group transfer estimates (EGTE).

For inference, in particular for calculating confidence intervals, we require the distribution of EGTE. For that purpose we rely on a block bootstrap procedure of the original time series [3,28]. We randomly draw blocks from the matrix of the original group of the considered four processes prior to discretization. The block length l_b is chosen following [29] as $T^{1/5}$ (rounded to the nearest integer), where T is the number of time series observations. Hence, the bootstrap proceeds as follows:

1. Randomly draw $n = T/l_b$ blocks of length l_b from the original matrix of time series;
2. Concatenate those n blocks to a bootstrapped time series of length T ; if T is not a multiple of l_b , the last block is of length $T - n \cdot l_b$;
3. Encode data via Equation (3) and estimate GTE according to Eq. (2);
4. Subtract the estimate of the bias to calculate EGTE;

Repeating these steps results in an estimate of the distribution of EGTE, which can be used to calculate confidence intervals, while the standard deviation of the bootstrapped EGTEs serves as an estimate for the EGTE standard error.

3. Simulation: Vector autoregression vs group transfer entropy

To unveil the benefits of GTE in comparison to standard methods in economics and finance, we conduct a simulation study using three time series (X_t, Y_t, Z_t). We incorporate structural breaks and nonlinear dependencies. The time series X_t is subject to random structural breaks, which correspond to a shift in the variance. The data generating processes are given by:

$$x_t = \begin{cases} \varepsilon_t & \text{with } \varepsilon_t \sim N(0, \sqrt{2}) \text{ for } t \leq T_1 \\ \varepsilon_t & \text{with } \varepsilon_t \sim N(0, 1) \text{ for } T_1 < t \leq T_2 \\ \varepsilon_t & \text{with } \varepsilon_t \sim N(0, \sqrt{1.5}) \text{ for } t > T_2 \end{cases}$$

$$y_t = 2\sqrt{|x_{t-1}|} + u_t, \quad \text{with } u_t \sim N(0, 1)$$

$$z_t = 0.25y_{t-1} + v_t, \quad \text{with } v_t \sim N(0, 1)$$

where T_1 and T_2 are chosen randomly in between $[T/4; 3T/4]$ and $T = 5000$ is the number of observations.

The main tools for assessing information flows in economics and finance are VARs and corresponding Granger-causality tests. Consequently, we use the simulated data to estimate a VAR(2) according to

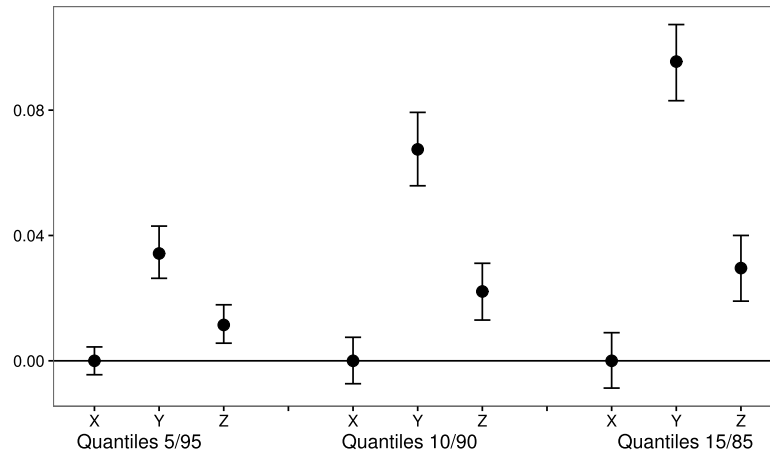
$$k_t = c + \Gamma_1 k_{t-1} + \Gamma_2 k_{t-2} + e_t \quad (4)$$

where $k_t = (x_t, y_t, z_t)'$ denotes the vector of simulated time series, c is a vector of constants, Γ_1 and Γ_2 denote a (3×3) matrix of the first and second order autoregressive parameters, respectively, and e_t denotes the vector of residuals. We use 500 simulation runs and test each simulated system for Granger-causality based on Eq. (4). We test the null hypothesis that the coefficients on first and second lags are jointly not statistically significant (excluding the variables own lag). Table 1 reports the percentage of simulated systems for which the null hypothesis of no Granger-causality is rejected on a 5% significance level.

The results show that the linear relationship between z_t and y_t leads to rejection of the null hypothesis of no Granger-causality in all simulation runs (last column) as the dependency between z_t and y_t is purely linear. The results show that the nonlinear relationship between y_t and x_t is not clearly detected by the Granger-causality tests as the null hypothesis is rejected for only 9.7% of the simulation runs. Consequently, the results might be misleading assuming that y_t cannot be predicted using lagged values of the other two variables. We then calculate EGTE estimates, using $k = l = 2$

Table 1Granger-causality tests (H_0 : variable not Granger-caused by other variables).

Variable	X	Y	Z
Cases H_0 rejected ($\alpha = 5\%$)	5.9%	9.7%	100%

**Fig. 1.** EGTE estimates for different quantile choices. The predicted variable is indicated on the x-axis.**Table 2**

Descriptive statistics and unit root test results.

	Mean	SD	Min	Max	Skew	Kurt	ADF	BH
BTC	0.000	0.017	−0.205	0.145	−0.232	16.940	−19.008	0.016
ETH	−0.000	0.021	−0.264	0.191	−0.075	18.278	−18.757	0.024
LTC	0.000	0.024	−0.292	0.331	0.635	27.579	−18.366	0.024
XRP	0.000	0.029	−0.388	0.331	0.416	30.401	−17.647	0.032

lags and three different quantile pairs ((0.05, 0.95), (0.10, 0.90), and (0.15, 0.85)) for the symbolic encoding in Eq. (3). The results are presented in Fig. 1. The figure shows that EGTE reveals the dependency of y_t on lagged values of x_t , as $EGTE_{(X,Z) \rightarrow Y}$ is significantly larger than zero. This is in contrast to the Granger-causality tests presented in Table 1, which did not unanimously support the hypothesis that Y is Granger-caused by the two other variables. The independence of X_t (conditional on its own lags) is also properly determined as $EGTE_{(Y,Z) \rightarrow X}$ estimates are not statistically significant. The structural breaks in X_t do not qualitatively alter the results. They also hold if T is varied. In the latter case, only the estimation error is affected in the expected way that the 90% confidence intervals increase with decreasing sample size.

4. Information flow between different cryptocurrencies

4.1. Data description

To illustrate the use of EGTE in finance, we examine the information flow between four cryptocurrency time series. We use transaction prices of Bitcoin (BTC), Ethereum (ETH), Litecoin (LTC), and Ripple (XRP) traded against the US Dollar at Kraken. The data range from 1 July 2017 to 2 October 2018, synchronously sampled at 1 h intervals, which results in 10,982 observations. They are obtained from CRYPTODATA.¹ Fig. 2 presents a time series plot of all entities. It shows the peak of Bitcoin (black solid line) on 17 December 2018 where its price reached almost 20,000 USD. The peak of LTC (black dotted line) occurs only two days later on 19 December 2017. The peaks of ETH (blue solid line) and XRP (blue dotted line, multiplied by 100) occurred early January 2018. Since then, all currencies dramatically lost value.

Table 2 provides selected descriptive statistics for the one hour cryptocurrency return time series. The minimum and maximum values underline the rise and fall of the cryptocurrencies. Their distributional behavior varies as BTC and ETH are negatively skewed while LTC and XRP exhibit positive skewness. All time series exhibit decisive excess kurtosis, albeit at varying degrees.

For EGTE estimation, the data have to be stationary. Since financial price series are commonly considered to contain a unit root, we calculate log returns as the first differences of the logarithm of transaction prices. To check whether the returns are stationary we use the Augmented Dickey–Fuller test [ADF, 30] and the unit root test of Buseti and Harvey [BH, 31]. The

¹ <https://www.cryptodatadownload.com/>.

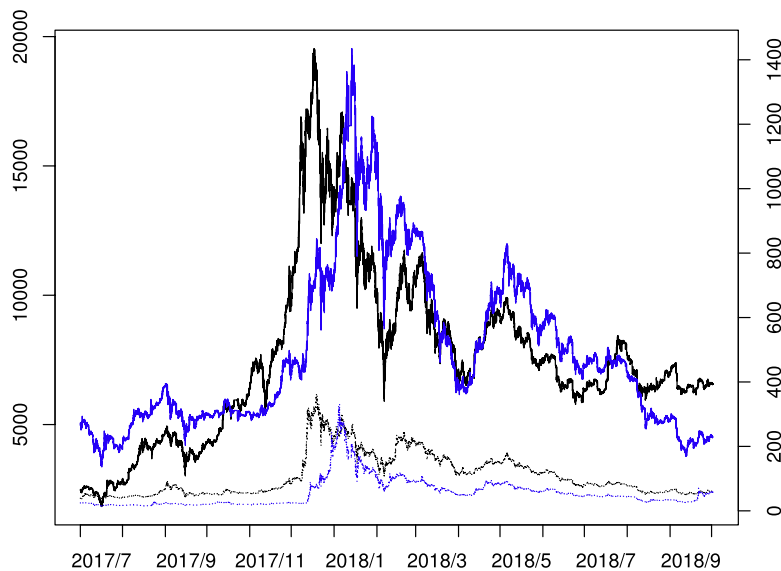


Fig. 2. Time series plots of BTC (black solid line, left axis), ETH (blue solid line), LTC (black dotted line), and XRP (blue dotted line, right axis, XRP multiplied by 100).

null hypothesis of the former is that the data contain a unit root while the null hypothesis of the latter is that the data do not contain a unit root. In all cases, the ADF null hypothesis that the returns contain a unit root is rejected (see column ADF in Table 2 for test statistics, the 1% critical value is -3.43). The BH test allows to additionally account for unknown structural breaks. We consider their Model 2 which allows for a break in the slope and the level, motivated by the pattern of prices observed in Fig. 2. In this case, the null hypothesis that the data do not contain a unit root cannot be rejected on any conventional significance level (see column BH in Table 2 for test statistics, the 10% critical value is 0.033).

4.2. Information flow between cryptocurrencies

Considering the choice of the quantiles used for discretization, Kernel density plots (as presented for Ethereum in Fig. 3) provide a suitable guideline. The plot compares the empirical distribution of Ethereum returns (solid line) to a normal distribution (dotted line) with the same mean and variance. The returns are much more condensed around the mean than a normal distribution. In a model for security prices like [32] or [33], returns are interpreted as the new information contribution to the fundamental price. Returns close to zero are therefore considered less informative than large (positive or negative) returns. Hence, the important, price relevant information lies in the tails. The plot depicts the 5% (95%) quantile of returns (filled dot), which is clearly higher (lower) than the quantile implied by a normal distribution (diamond). Hence, using quantiles between 5%–15% for the first and 85%–95% for the third bin and is a sensible choice for the discretization.

We estimate EGTE using three bins and alter the quantile pairs for discretization of the data ((0.05, 0.95), (0.10, 0.90), and (0.15, 0.85)). Results are summarized in Fig. 4, which presents point estimates of EGTE together with 90% confidence bounds. The graph shows statistically significant EGTE estimates for all four cryptocurrencies for any of the three sets of quantiles used for discretization. Hence, there is information flow towards any of the cryptocurrencies from the set of remaining ones.

Considering the results using the (0.05, 0.95) quantile pair, we find that the information flow towards Bitcoin is higher than towards Ethereum, Litecoin, or Ripple. However, the differences are small, the confidence bounds overlap so that the hypothesis that there is equal information flow across all cryptocurrencies cannot be rejected. This conclusion is supported by the results using the (0.10, 0.90) quantile pair for which the equality is even more pronounced as illustrated in Fig. 4. While we still cannot reject the hypothesis when using the (0.15, 0.85) quantile pair, the differences here are slightly more explicit: there is less information flow towards BTC and LTC than towards ETH and XRP. The fact that these pairs behave similar is in line with Fig. 2 which shows that BTC and LTC follow a similar up-and-down pattern across the considered time period (as do ETH and XRP for that matter).

As the time period considered covers the strong bull market of cryptocurrencies which ended mid/end December 2017 (depending on which currency is taken as a proxy) and a (lasting) bear market which basically covers the entire period of 2018, we split the sample and conduct the analysis on subperiods 1 July 2017 to 17 December 2017 (bull market sample) and 18 December 2017 to 2 October 2018 (bear market sample). The results are summarized in Table 3 based on a discretization of the data using the (0.05, 0.95) quantile pair (the results marked “Full” correspond to the ones presented in Fig. 4). The subsample analysis reveals that the overall information transmission in the full sample is dominated by the bear market. The order of magnitude of the EGTE estimates is very similar in the full sample and the bear market sample. In the bull

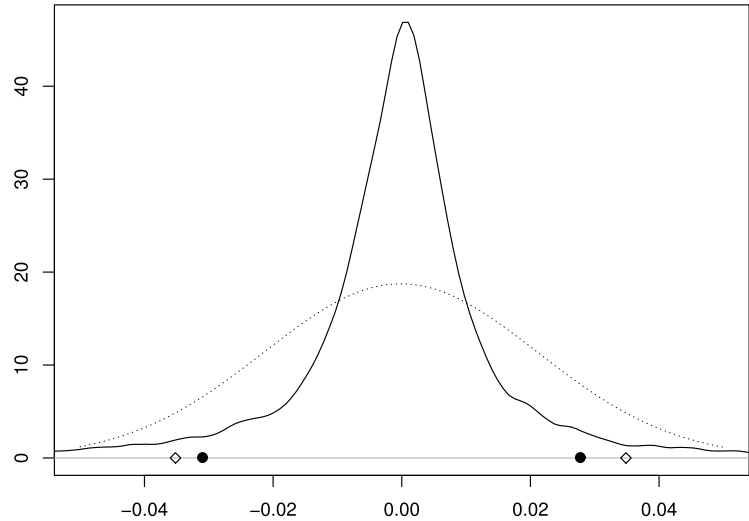


Fig. 3. Density plot of Ethereum returns.

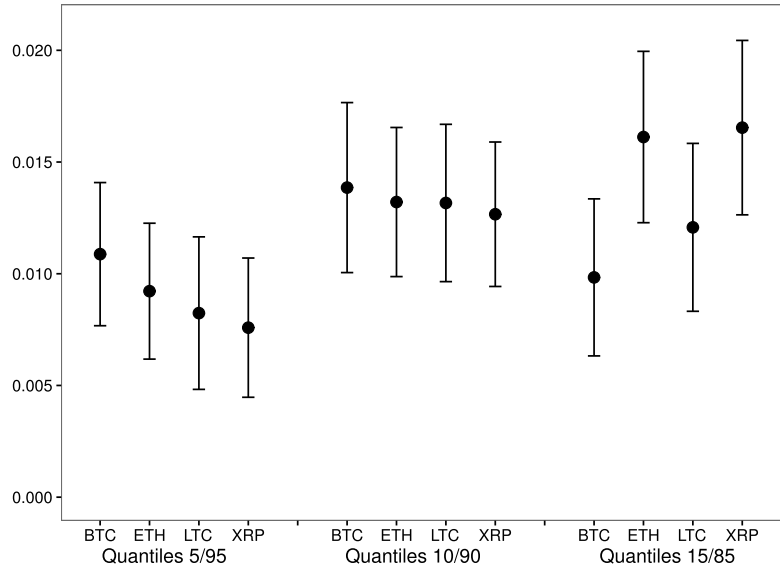


Fig. 4. Cryptocurrency EGTE estimates for different quantile choices. The predicted cryptocurrency is indicated on the x-axis.

market sample, the EGTE estimates are generally higher which indicates that in this period, predictability of any of the cryptocurrencies has been higher as in the overall sample. Hence, uncertainty about any of the cryptocurrencies could be better reduced using the group of other cryptocurrencies than in the current bear market situation.

Our sample time period also contains three potentially important Bitcoin forks: Bitcoin Cash (1 August 2017), Bitcoin Gold (24 October 2017), and Bitcoin Private (28 February 2018). In the simulation study in Section 3 we showed that the calculation of EGTE is not affected by single outliers. To establish that this conjecture also holds empirically, we calculate EGTE for the full sample, excluding the 2 days before and after the forks and the day of the fork itself in the calculation of the empirical frequencies.² It turns out that the EGTE estimates are qualitatively not affected. Indeed, the order of magnitude is similar to the one reported in Table 3. This illustrates the strength of transfer entropy as a robust measure for information flow which is qualitatively not affected by single extreme events.

² Detailed results are available from the corresponding author upon request.

Table 3

Effective group transfer entropy (EGTE) between cryptocurrencies with standard errors and p -values for different market conditions.

Sample		BTC	ETH	LTC	XRP
Full	EGTE	0.01088	0.00922	0.00824	0.00759
	se(EGTE)	0.00195	0.00185	0.00208	0.00189
	p -value	0.00000	0.00000	0.00007	0.00006
Bull market	EGTE	0.01497	0.01742	0.01058	0.01843
	se(EGTE)	0.00359	0.00427	0.00396	0.00421
	p -value	0.00003	0.00005	0.00749	0.00001
Bear market	EGTE	0.00994	0.00922	0.00840	0.01059
	se(EGTE)	0.00281	0.00251	0.00264	0.00272
	p -value	0.00040	0.00024	0.00147	0.00010

Table 4

Granger-causality tests (H_0 : variable in column is not Granger-caused by other variables).

Sample		BTC	ETH	LTC	XRP
Full	test stat.	1.37165	2.05574	2.37525	1.23178
	p -value	0.19460	0.02980	0.01109	0.26990
Bull market	test stat.	1.32237	1.37249	2.11736	0.52783
	p -value	0.24303	0.22147	0.04804	0.78761
Bear market	test stat.	1.80751	1.55728	2.15153	1.53168
	p -value	0.06155	0.12184	0.02230	0.13024

4.3. VAR approach and EGTE using VAR residuals

To compare our results to the standard approach used in financial applications, we estimate a VAR model and conduct Granger-causality tests. The lag order is selected using the Schwartz Bayesian information criterion (BIC, [34]). We set up a VAR(3) for the full sample period as

$$r_t = c + \Gamma_1 r_{t-1} + \Gamma_2 r_{t-2} + \Gamma_3 r_{t-3} + u_t \quad (5)$$

where $r_t = (r_t^{XMR}, r_t^{BTC}, r_t^{ETH}, r_t^{LTC})'$ denotes the vector of returns, c is a (4×1) vector of constants, Γ_i , $i \in \{1, 2, 3\}$, denotes a (4×4) matrix of the first order autoregressive parameters, and u_t denotes the vector of serially uncorrelated innovations. Granger-causality tests are then performed on the autoregressive parameters. We test the null hypothesis that the coefficients on the included lags are jointly not statistically significant (excluding the variables own lags). The Granger-causality test results are reported in Table 4. In the full sample period, the corresponding p -values indicate a rejection of the null hypothesis of no Granger-causality for Ethereum and Litecoin on a 5% significance level, while the results for Bitcoin and Ripple show that the null hypothesis cannot be rejected. For investors this implies that current values of Bitcoin and Ripple cannot be predicted using lagged observations of the other variables on an hourly basis. A trading strategy that involves Ethereum and Litecoin might, on the other hand, exploit the weak Granger-causality that is documented for these cryptocurrencies. These results are only partially in line with our EGTE estimates documented in Table 3. EGTE indicated the highest information flow towards Bitcoin and the lowest towards Ripple. While the latter is not Granger-caused, Bitcoin is not either. This might be a hint that there is more in the dynamics than the linear model can grasp.

To ensure that the results are not driven by the sample period, we also conduct the VAR analysis on the bull and bear market samples. In both cases the BIC suggests a VAR(2). Overall, the results are similar. In the bull market, no cryptocurrency but Litecoin is Granger-caused by the others (on a 5% significance level). Again, this seems counter-intuitive as during that sample period the markets only seemed to know one direction: upwards. However, on the hourly basis of our analysis this does not lead to predictability. In the bear market, Bitcoin (on a 10% significance level) and Litecoin (on a 5% significance level) are Granger-caused by the respective set of other cryptocurrencies. This seems plausible if we consider again the time series plot in Fig. 2 which illustrates that BTC and LTC exhibit a very closely related sample path (albeit the different levels). Considering the parameter estimates, the in the models of BTC and LTC, the respective other variable turns out statistically significant on the first lag which also drives the results of the Granger-causality tests. All in all, predictability in a Granger-causal sense is rather weak and depends on the significance level chosen. As we have more than 10,000 observations available one might be inclined to select a stricter one than 5% in which case the hypothesis that there is no predictability at all could not be rejected for any of the models.

These somewhat conflicting results might be due to the fact that Granger-causality tests are limited to linear dynamics. To further examine this issue, we re-estimate EGTE using the VAR residuals which result from estimation of Eq. (5) as input, i.e. $\hat{u}_t = r_t - \sum_i \hat{\Gamma}_i r_{t-i}$. These residual series are purged by the linear effects and significant EGTE estimates indicate nonlinear information flow beyond what is captured by the VAR model.

Table 5
EGTE on VAR residuals using quantiles (5/95).

Sample		BTC	ETH	LTC	XRP
Full	EGTE	0.00998	0.01359	0.00962	0.00646
	se(EGTE)	0.00234	0.00205	0.00207	0.00208
	p-value	0.00002	0.00000	0.00000	0.00187
Bull market	EGTE	0.01361	0.01636	0.01182	0.01329
	se(EGTE)	0.00520	0.00454	0.00496	0.00466
	p-value	0.00891	0.00031	0.01712	0.00439
Bear market	EGTE	0.01021	0.01289	0.00998	0.00683
	se(EGTE)	0.00298	0.00295	0.00301	0.00298
	p-value	0.00062	0.00001	0.00092	0.02174

The results using quantiles (0.05,0.95) for the symbolic encoding are presented in Table 5. The p -values in Table 5 indicate significant EGTE estimates for all cryptocurrencies.³ Consequently, there must be nonlinear information transfer not captured by the linear model in Eq. (5). The largest estimates are found for Monero and Ethereum, while Litecoin are the least affected by lagged observations of the other three variables. With respect to Bitcoin (BTC) and Ethereum (ETH), we can conclude that information transfer is completely nonlinear in nature, as the previous Granger-causality tests do not indicate a linear impact of the other three currencies.

As in the case when estimating EGTE on the data in Section 4.2, the Bitcoin forks did not turn out to qualitatively affect the results of the VAR analysis and the subsequent calculation of EGTE on the VAR residuals. To account for them when estimating the VAR, we included a dummy variable which is one on the day of the forks and zero else.⁴ In the estimation, the dummy variable never turns out to be statistically significant (with p -values decisively higher than 0.1). The conclusions drawn from the Granger-causality tests are therefore also not affected as is the subsequent transfer entropy analysis. Indeed, the order of magnitude of the EGTE estimates is very close to the estimates documented in Table 5.

5. Conclusion

We propose to use effective group transfer entropy (EGTE) to measure the dependence of a stochastic process on lagged values of other stochastic processes within a multivariate system. In particular within the area of economics and finance, standard approaches and tools are based on linear models such as vector autoregressive models and Granger-causality tests. Our simulation study shows that these approaches might result in misleading conclusions due to their restriction to linear relationships. EGTE offers an interesting alternative approach, as it is able to capture more general forms of statistical dependencies.

In our empirical analysis we examine intraday data on four different cryptocurrencies (Bitcoin, Ethereum, Litecoin, and Monero) and find that information transfer between cryptocurrency markets is mostly a nonlinear process, which emphasizes the importance of flexible measures for information flows, such as EGTE. We find that Bitcoin is not the one dominating cryptocurrency which leads the information process in all of these markets. While Bitcoin has probably received the most attention in the media during the last couple of years, this advantage of information dissemination does not translate into a dominant position in the context of price formation among the cryptocurrencies.

References

- [1] T. Schreiber, Measuring information transfer, *Phys. Rev. Lett.* 85 (2) (2000) 461–464.
- [2] M. Prokopenko, J.T. Lizier, D.C. Price, On thermodynamic interpretation of transfer entropy, *Entropy* 15 (2) (2013) 524–543.
- [3] T. Dimpfl, F.J. Peter, Using transfer entropy to measure information flows between financial markets, *Stud. Nonlinear Dyn. Econom.* 17 (1) (2013) 85–102.
- [4] T. Dimpfl, F.J. Peter, The impact of the financial crisis on transatlantic information flows: An intraday analysis, *J. Int. Financial Mark. Inst. Money* 31 (2014) 1–13.
- [5] L. Sandoval Junior, A. Mullokanov, D.Y. Kenett, Dependency relations among international stock market indices, *J. Risk Financial Manag.* 8 (2) (2015) 227–265.
- [6] A.G. Dimitrov, A.A. Lazar, J.D. Victor, Information theory in neuroscience, *J. Comput. Neurosci.* 30 (1) (2011) 1–5.
- [7] P.-O. Amblard, O.J. Michel, On directed information theory and granger causality graphs, *J. Comput. Neurosci.* 30 (1) (2011) 7–16.
- [8] M. Wibral, R. Vicente, M. Lindner (Eds.), *Directed Information Measures in Neuroscience*, Springer, 2014.
- [9] J. Lee, S. Nemati, I. Silva, B.A. Edwards, J.P. Butler, A. Malhotra, Transfer entropy estimation and directional coupling change detection in biomedical time series, *Biomed. Eng. online* 11 (1) (2012) 19.
- [10] L. Zheng, W. Pan, Y. Li, D. Luo, Q. Wang, G. Liu, Use of mutual information and transfer entropy to assess interaction between parasympathetic and sympathetic activities of nervous system from hrv, *Entropy* 19 (2017) 489.
- [11] M. Oh, S. Kim, K. Lim, S.Y. Kim, Time series analysis of the antarctic circumpolar wave via symbolic transfer entropy, *Physica A* 499 (2018) 233–240.

³ Results based on the (0.10, 0.90) and (0.15, 0.85) quantiles for discretization of the data are lead to the same conclusions. Details are available upon request.

⁴ Detailed results are available from the corresponding author upon request.

- [12] J. Hasbrouck, One security many markets: Determining the contributions to price discovery, *J. Finance* 50 (4) (1995) 1175–1199.
- [13] D. Lien, K. Shrestha, A new information share measure, *J. Futures Mark* 29 (4) (2009) 377–395.
- [14] J. Grammig, F.J. Peter, Telltale tails: A new approach to estimating unique market information shares, *J. Financial Quant. Anal.* 48 (2) (2013) 459–488.
- [15] T.J. Putnig, What do price discovery metrics really measure? *J. Empir. Finance* 23 (2013) 68–83, <http://dx.doi.org/10.1016/j.jempfin.2013.05.004>, URL: <http://www.sciencedirect.com/science/article/pii/S0927539813000340>.
- [16] J. Grammig, M. Melvin, C. Schlag, Internationally cross-listed stock prices during overlapping trading hours: Price discovery and exchange rate effects, *J. Empir. Finance* 12 (1) (2005) 139–164.
- [17] S. Dolatabadi, M.O. Nielsen, K. Xu, A fractionally cointegrated VAR analysis of price discovery in commodity futures markets, *J. Futures Mark* 35 (4) (2015) 339–356.
- [18] J.M. Nichols, M. Seaver, S.T. Trickey, M.D. Todd, C. Olson, L. Overbey, Detecting nonlinearity in structural systems using the transfer entropy, *Phys. Rev. E* 72 (4) (2005) 04–17.
- [19] L. Barnett, A.B. Barrett, A.K. Seth, Granger causality and transfer entropy are equivalent for gaussian variables, *Phys. Rev. Lett.* 103 (23) (2009) 238701.
- [20] L. Kristoufek, On bitcoin markets (in)efficiency and its evolution, *Physica A* 503 (2018) 257–262, <http://dx.doi.org/10.1016/j.physa.2018.02.161> URL: <http://www.sciencedirect.com/science/article/pii/S0378437118302413>.
- [21] W.C. Wei, Liquidity and market efficiency in cryptocurrencies, *Econom. Lett.* 168 (2018) 21–24, <http://dx.doi.org/10.1016/j.econlet.2018.04.003> URL: <http://www.sciencedirect.com/science/article/pii/S0165176518301320>.
- [22] A. Urquhart, The inefficiency of bitcoin, *Econom. Lett.* 148 (2016) 80–82, <http://dx.doi.org/10.1016/j.econlet.2016.09.019>, URL: <http://www.sciencedirect.com/science/article/pii/S0165176516303640>.
- [23] A. Brauneis, R. Mestel, Price discovery of cryptocurrencies: bitcoin and beyond, *Econom. Lett.* 165 (2018) 58–61, <http://dx.doi.org/10.1016/j.econlet.2018.02.001>, URL: <http://www.sciencedirect.com/science/article/pii/S0165176518300417>.
- [24] M. Staniek, K. Lehnertz, Symbolic transfer entropy, *Phys. Rev. Lett.* 100 (15) (2008) 158101.
- [25] M.S. Daugherty, T. Jithendranathan, A study of linkages between frontier markets and the u.s. equity markets using multivariate garch and transfer entropy, *J. Multinational Financial Manag.* 32–33 (2015) 95–115, <http://dx.doi.org/10.1016/j.mulfin.2015.10.003>, URL: <http://www.sciencedirect.com/science/article/pii/S1042444X15300086>.
- [26] F. Longin, B. Solnik, Extreme correlation of international equity markets, *J. Finance* 56 (2001) 649–676.
- [27] R. Marschinski, H. Kantz, Analysing the information flow between financial time series: An improved estimator for transfer entropy, *Eur. Phys. J. B* 30 (2) (2002) 275–281.
- [28] J.G. MacKinnon, Bootstrap methods in econometrics, *Econ. Rec.* 82 (s1) (2006) 2–18.
- [29] P. Hall, J.L. Horowitz, B.-Y. Jing, On blocking rules for the bootstrap with dependent data, *Biometrika* 82 (3) (1995) 561–574.
- [30] D. Dickey, W.A. Fuller, Likelihood ratio statistics for autoregressive time series with a unit root, *Econometrica* 49 (4) (1981) 1057–1072.
- [31] F. Busetti, A. Harvey, Testing for the presence of a random walk in series with structural breaks, *J. Time Series Anal.* 22 (2) (2001) 127–150.
- [32] R. Roll, A simple implicit measure of the effective bid–ask spread in an efficient market, *J. Finance* 39 (4) (1984) 1127–1139.
- [33] A. Madhavan, M. Richardson, M. Roomans, Why do security prices change? A transaction-level analysis of nyse stocks, *Rev. Financ. Stud.* 10 (4) (1997) 1035–1064.
- [34] G. Schwarz, Estimating the dimension of a model, *Ann. Statist.* 6 (2) (1978) 461–464.